

This command provides 3 lines of output over a 1-second delay:

```
-----cpu-----
24
225
339
```

The first line gives the average number of context switches over 1 second since the system booted, and the next two lines give the number of context switches over two 1-second intervals. Since this machine booted, it has averaged 24 context switches per second. And in the past second, 225 context switches were made, with 339 context switches in the second prior to that.

We can also use the `/proc` file system to determine the number of context switches for a given process. For example, the contents of the file `/proc/2166/status` will list various statistics for the process with `pid = 2166`. The command

```
cat /proc/2166/status
```

provides the following trimmed output:

```
voluntary_ctxt_switches      150
nonvoluntary_ctxt_switches   8
```

This output shows the number of context switches over the lifetime of the process. Notice the distinction between *voluntary* and *nonvoluntary* context switches. A voluntary context switch occurs when a process has given up control of the CPU because it requires a resource that is currently unavailable (such as blocking for I/O.) A nonvoluntary context switch occurs when the CPU has been taken away from a process, such as when its time slice has expired or it has been preempted by a higher-priority process.

5.2 Scheduling Criteria

Different CPU-scheduling algorithms have different properties, and the choice of a particular algorithm may favor one class of processes over another. In choosing which algorithm to use in a particular situation, we must consider the properties of the various algorithms.

Many criteria have been suggested for comparing CPU-scheduling algorithms. Which characteristics are used for comparison can make a substantial difference in which algorithm is judged to be best. The criteria include the following:

- **CPU utilization.** We want to keep the CPU as busy as possible. Conceptually, CPU utilization can range from 0 to 100 percent. In a real system, it should range from 40 percent (for a lightly loaded system) to 90 percent (for a heavily loaded system). (CPU utilization can be obtained by using the `top` command on Linux, macOS, and UNIX systems.)
- **Throughput.** If the CPU is busy executing processes, then work is being done. One measure of work is the number of processes that are completed

per time unit, called **throughput**. For long processes, this rate may be one process over several seconds; for short transactions, it may be tens of processes per second.

- **Turnaround time.** From the point of view of a particular process, the important criterion is **how long it takes to execute that process**. The interval from the **time of submission of a process to the time of completion** is the turnaround time. Turnaround time is the **sum of the periods spent waiting in the ready queue, executing on the CPU, and doing I/O**.
- **Waiting time.** The CPU-scheduling algorithm does not affect the amount of time during which a process executes or does I/O. It **affects only** the amount of time that **a process spends waiting in the ready queue**. Waiting time is the sum of the periods spent waiting in the ready queue.
- **Response time.** In an **interactive system**, **turnaround time may not be the best criterion**. Often, a process can produce some output fairly early and can continue computing new results while previous results are being output to the user. Thus, another measure is **the time from the submission of a request until the first response is produced**. This measure, called response time, is the **time it takes to start responding**, not the time it takes to output the response.

It is **desirable to maximize CPU utilization** and **throughput** and **to minimize turnaround time, waiting time, and response time**. In most cases, we optimize the average measure. However, under some circumstances, we prefer to optimize the minimum or maximum values rather than the average. For example, to guarantee that all users get good service, we may want to minimize the maximum response time.

Investigators have suggested that, for **interactive systems** (such as a PC desktop or laptop system), it is more **important to minimize the variance in the response time than to minimize the average response time**. **A system with reasonable and predictable response time** may be **considered more desirable than a system that is faster on the average but is highly variable**. However, little **work has been done** on **CPU-scheduling algorithms** that **minimize variance**.

As we discuss various CPU-scheduling algorithms in the following section, we illustrate their operation. An accurate illustration should involve many processes, each a sequence of several hundred CPU bursts and I/O bursts. For simplicity, though, we consider only one CPU burst (in milliseconds) per process in our examples. Our measure of comparison is the average waiting time. More elaborate evaluation mechanisms are discussed in Section 5.8.

5.3 Scheduling Algorithms

CPU scheduling deals with the problem of deciding which of the processes in the ready queue is to be allocated the CPU's core. There are many different CPU-scheduling algorithms. In this section, we describe several of them. Although most modern CPU architectures have multiple processing cores, we describe these scheduling algorithms in the context of only one processing core available. That is, a single CPU that has a single processing core, thus the system is